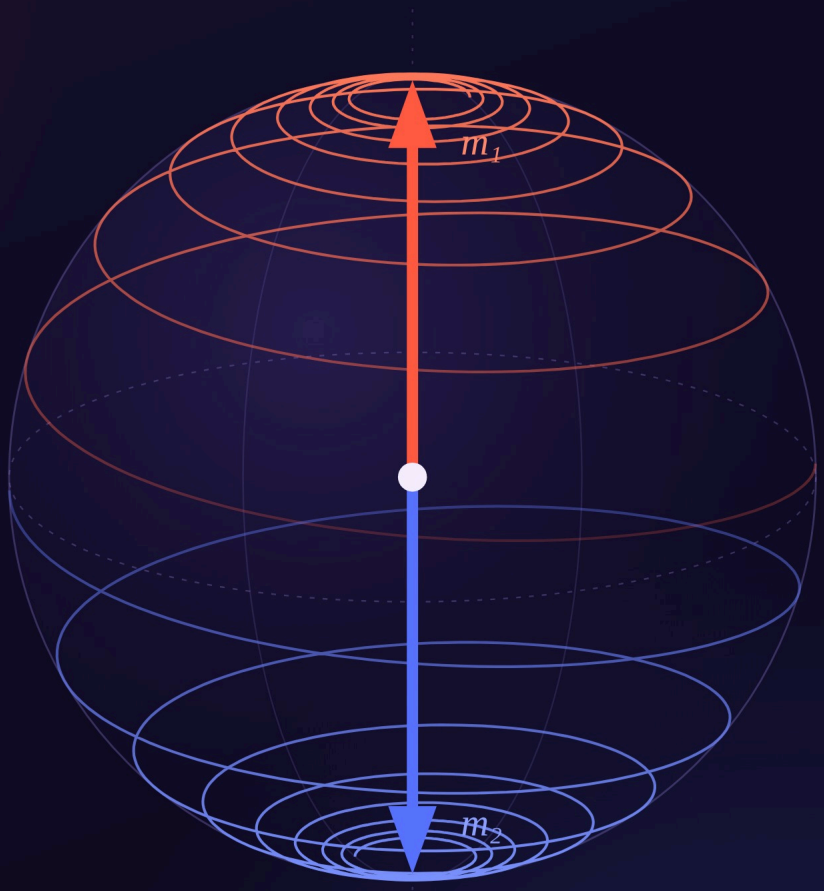
spinmate



USER GUIDE

Spintronics simulation, by conversation.

An AI agent that validates your setup with the solver's own parser, runs two-sublattice macrospin simulations and explains the results from the actual magnetization dynamics, with figures and reports ready to share.

Validated solvers
Macrospin LLG on 9 analytic cases; 2D micromagnetics for skyrmions; mumax3 in

Claude, OpenAI, Gemini
Routed per request by capability and cost

Any agent (MCP)
Web app, Claude Code, Claude Desktop

Runs locally
Docker or Windows app; your keys, your data

Hermes · Dionysus · Zeus · Apollo · Athena

spinmate

User Guide

Version 2.0 · free and open source (MIT) · github.com/Michaellianeris/Spinmate-AI-Agent

An AI agent for spintronics simulation. Describe the setup in plain language or provide a parameter file. SpinMate validates the input with the solver's own parser, runs two-sublattice macrospin simulations, and explains the results from the actual magnetization dynamics, with figures and reports ready to share.

At a glance

- Chat-driven runs, sweeps and threshold searches on the two-sublattice macrospin (LLG) solver
- Answers grounded in computed metrics: switches, settling, frequencies
- **New in 2.0:** an MCP server, so Claude Code, Claude Desktop or any agent uses the same tools
- **New in 2.0:** a physics validation suite against analytic results, SpinMate's own built-in 2D micromagnetic engine (skyrmions, domain walls) and a mumax3 engine plug-in
- Claude, OpenAI and Gemini behind four personas, chosen per request by capability and price
- Figures (IEEE, Nature), reports with LaTeX math, a searchable run catalog
- Six interface languages; runs locally with Docker or a Windows app

Contents

1. Getting started
2. Your first simulation
3. Reading a run card
4. Fast, Deep and next steps
5. Models and personas
6. The tools
7. Architecture
8. From Claude and other agents
9. Engines: macrospin, mm2d and mumax3
10. Physics validation
11. Sweeps and continuing runs
12. Compare, figures, reports
13. Runs panel and catalog
14. Settings and conversations
15. Data, privacy and security
16. Configuration
17. Parameters
18. Troubleshooting

1 Getting started

Docker (Linux, macOS, Windows/WSL)

```
cp .env.example .env      # add ANTHROPIC_API_KEY, OPENAI_API_KEY and/or GEMINI_API_KEY
docker compose up -d      # then open http://localhost:8421
```

Windows desktop app

Run `SpinMate.exe`; it starts the server and opens the browser. Runs are kept in `%LOCALAPPDATA%\SpinMate\data` and the catalog is SQLite.

First start

1. **Athena**, the owl, asks your name and, optionally, your email. The email creates an account and, if a mail server is set up, brings this guide as a welcome email; otherwise she gives you the download link.
2. **API key:** Settings (gear) → paste an Anthropic, OpenAI and/or Gemini key (Gemini keys: aistudio.google.com/api-keys). A pasted key stays in that browser tab; a key in `.env` serves everyone on the server.
3. **Ask** in the message box, or pick a card under *Find a skill*.

No key? *Run setup* (play icon) runs a preset or your parameter file directly; only the chat needs a model.

2 Your first simulation

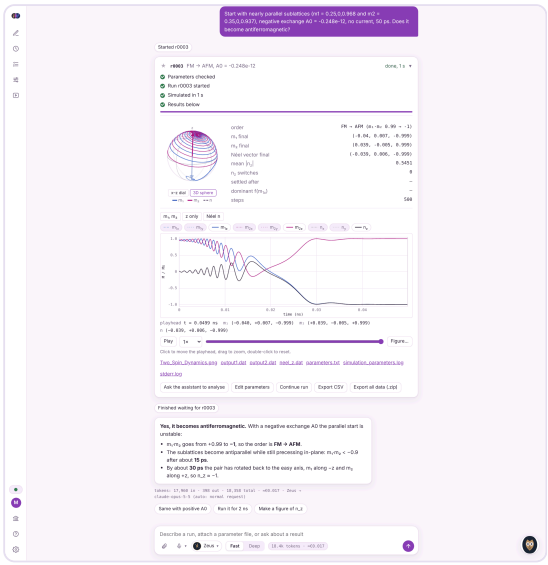
Describe the starting state, exchange, fields, current or temperature, and the duration (SI units, currents in A/cm²).

> *Start with nearly parallel sublattices (m1 = 0.25,0,0.968 and m2 = 0.35,0,0.937), A0 = −0.248e-12, 50 ps. Does it become antiferromagnetic?*

> *Start antiferromagnetic with a positive exchange A0 = 0.248e-12, 120 ps. Does it end ferromagnetic?*

> *Apply a DC spin-orbit current of 100e6 A/cm² for 2 ns to the default AFM state. What happens?*

> *Heat a FeRh layer through its transition: positive A0, 370 K ± 60 K at 0.25 GHz, 2 ns.*



Question, run card and answer with next steps.

The parameters are checked with the solver's own parser before anything runs; a suspicious value (for example `A0 = 0.248` without `e-12`) is pointed out instead of run. Large requests (many runs or long runtimes) wait for your **Run it** button. The answer ends with suggested next steps as buttons.

What happens behind one question

1. The router picks a model for the persona and the request.
2. The model calls `validate_parameters`, then `start_simulation`.
3. The run queue starts the solver in its own process; the card shows live progress.
4. `wait_for_run` returns the summary metrics; the model writes the answer from them.

3 Reading a run card

Field	Meaning
order	Initial and final order from $m_1 \cdot m_2$: AFM below −0.9, FM above +0.9, otherwise canted.
m_1, m_2, Néel final	Final unit vectors and the Néel vector $n = (m_1 - m_2)/2$.
n_z switches	Real reversals of n_z (from below −0.5 to above +0.5 or back) and the time of the first; ringing around zero is not counted.
settled after	When m_1 and m_2 stop changing within 2 %; "–" if they did not settle within the run.
dominant f	Main frequency of m_{1x}, reported only if at least five periods fit in the second half of the run.

The **sphere** shows the trajectories of m_1 (blue) and m_2 (magenta) in 3D or as an x–z dial. The **chart** shows m_1/m_2, z-components only, or the Néel vector: drag to zoom, double-click to reset, click to move the playhead; **Play** animates the run. The card links the raw files and exports CSV or a zip of all data.

4 Fast, Deep and next steps

Fast uses the default parameters, changes only what you name and answers in a few lines. **Deep** works from the full parameter set or your file, states the caveats that apply and gives a quantitative explanation. Both end with up to three suggested follow-ups. Requests estimated above `CONFIRM_MINUTES` or `CONFIRM_RUNS` pause for **Run it**.

5 Models and personas

Under the message box you pick a persona, not a model. Each persona groups models of one level from all three providers.

Persona	Best for	Typical models
Hermes	Single runs, presets, short answers	Claude Haiku · GPT nano / mini · Gemini Flash-Lite
Dionysus (default)	Everyday runs with a clear explanation, sweeps, comparisons	Claude Sonnet · GPT-5.x · Gemini Flash
Zeus	Deep mode, goal searches, careful curve reading, reports	Claude Opus · OpenAI o-series · Gemini Pro
Apollo	New or special models as they appear	Claude Fable / Mythos · new Gemini lines · new models
Athena	The how-to guide: where things are, what buttons do	Selected provider · built-in answers offline

How the model is chosen

On *Auto* each request is scored for every model of the persona whose key is present:

```
score = capability + bonus(strongest of the persona) − λ(persona, request) × log10(price / cheapest) − shortfall
```

Capability comes from the model family and generation; price from list prices; λ is largest for Hermes and smallest for Zeus and shrinks for demanding requests (Deep mode, sweeps, comparisons, reports, long messages). So the strongest model wins unless a cheaper one is clearly enough. There is no fixed provider order; on an API error the next model by score is used. Pro models run only when pinned (Settings → *Model behind each god*). Every answer shows the model, the reason and the cost in €, converted with the daily ECB reference rate.

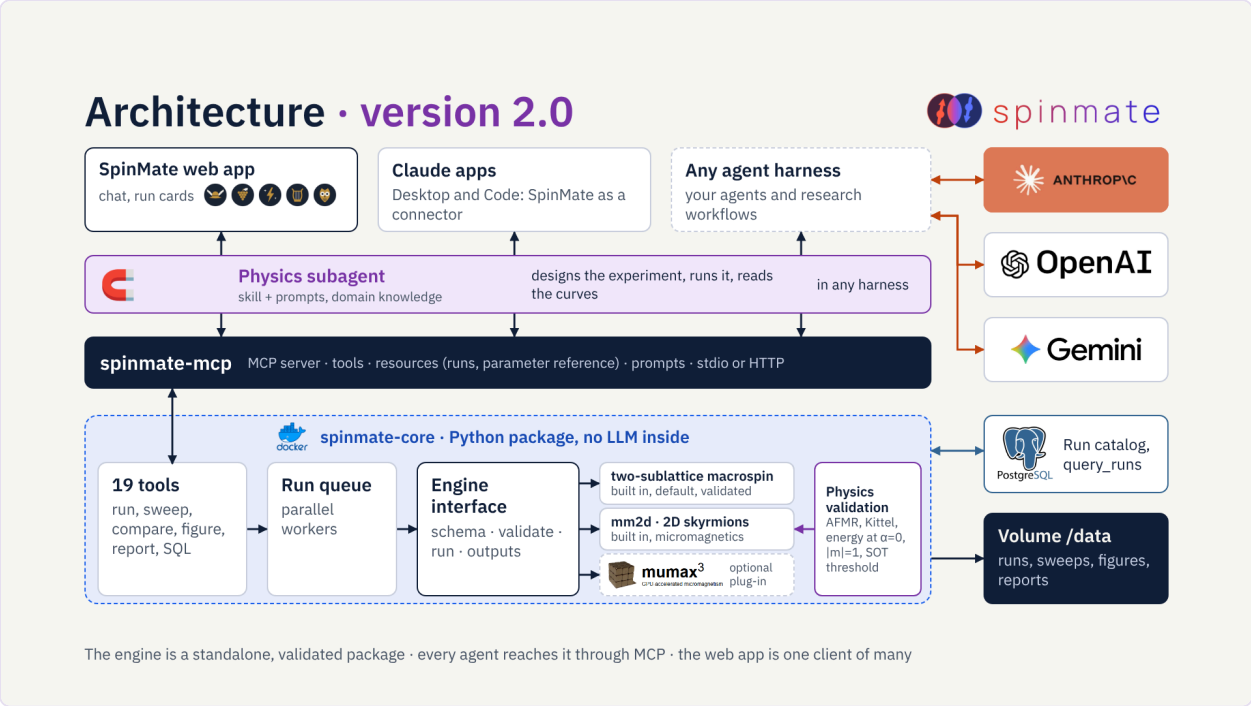
6 The tools

Group	Tools
Inputs	`list_input_files`, `read_input_file`, `save_input_file`, `validate_parameters`, `list_engines`
Runs	`start_simulation`, `wait_for_run`, `get_run`, `list_runs`, `cancel_run`
Sweeps	`start_sweep`, `wait_for_sweep`, `get_sweep`

Continuation	`continue_run`, `combine_runs`
Analysis & output	`compare_runs`, `make_figure`, `make_report`, `query_runs` (SQL on the catalog)

The same 19 tools serve the chat and every MCP client (chapter 8). `query_runs` appears only when the catalog is on. `start_simulation` and `start_sweep` stop large requests: in the chat they wait for **Run it**, over MCP they return `needs_confirmation` and run only when called again with `confirmed=true`.

7 Architecture



Version 2.0: every client reaches the same MCP server, on top of spinmate-core and its engines.

Component	Role
Browser (`web/index.html`)	Chat, run cards, sphere and charts, Runs panel, figure and report cards, voice input, six languages. Conversations are kept in the browser.
FastAPI server (`app/server.py`)	REST endpoints for runs, sweeps, figures, reports, accounts, transcription; streams the agent's events to the page; serves the MCP server at `/mcp`.
Agent loop (`app/agent.py`)	Provider-neutral tool loop for Claude, OpenAI and Gemini, with prompt caching, token budget and cost accounting. It calls the tools through MCP (`app/mcp_client.py`), like any other client.
Model router (`app/router.py`)	Scores the persona's models on capability and price for each request.
MCP server (`spinmate_mcp/`)	Tools, resources (parameters, physics guide, validation, runs, examples) and prompts, over stdio or HTTP. `physics.md` is the physics guide shared by the server, the chat and the Claude Code subagent.
spinmate-core (`spinmate_core/`)	The engine without an LLM: tools, run queue, sweeps and continuation, catalog, figures, reports, validation suite and a command line (`python -m spinmate_core`).
Engines (`spinmate_core/engines/`)	One interface: `parameter_schema`, `validate`, `run_command`, `outputs`. macrospin and mm2d are built in; plug-ins such as mumax3 register under `spinmate.engines`.

Run catalog	Postgres in Docker, SQLite in the desktop app: one row per run, queried with read-only SQL.
Data volume	`/data`: runs, sweeps, figures, reports, validation, accounts, exchange-rate cache.

8 From Claude and other agents (MCP)

SpinMate is also an MCP server (Model Context Protocol), so Claude Code, Claude Desktop or any MCP agent can design, run and read simulations with the same tools, runs and queue as the web app. The plug icon in the left bar shows these commands for your server, with copy buttons.

Claude Code

With the Docker app running:

```
claude mcp add --transport http spinmate
http://localhost:8421/mcp
```

The repository also brings a physics subagent (`.claude/agents/spinmate-physicist.md`) and a skill (`.claude/skills/spinmate-physics`), used automatically when you open Claude Code in the repository.

Without Docker

```
python -m spinmate_mcp          # stdio
python -m spinmate_mcp --http   # port 8422
```

Claude Desktop

Settings → Developer → Edit config (`claude_desktop_config.json`):

```
{"mcpServers": {"spinmate": {
  "command": "docker",
  "args": ["compose", "-f",
"/path/to/Spinmate-AI-Agent/docker-
compose.yml",
           "exec", "-T", "sublattice-agent",
"python", "-m", "spinmate_mcp"]}}}
```

Links in the results (figures, reports) point to `APP_URL`, by default `http://localhost:8421`.

The server offers	Contents
Tools	The 19 tools of chapter 6; `get_run` and `make_figure` return the plot as an image.
Resources	`spinmate://parameters`, `physics/guide`, `physics/scales`, `validation`, `runs`, `runs/{id}`, `runs/{id}/parameters.txt`, `examples/{name}`
Prompts	`design_experiment` (a question → runs → interpretation), `find_threshold` (coarse and fine sweeps), `interpret_run`

> *Use spinmate to find the smallest DC SOT current that makes the Néel vector rotate.*

9 Engines: macrospin, mm2d and mumax3

Two engines are built in: macrospin (two sublattices) and mm2d (2D micromagnetics: skyrmions, domain walls). mumax3 or any other engine is an optional plug-in, added only if you install it. Every run records the engine that produced it. `list_engines` shows what is installed and the parameters of each; in the chat simply ask for a skyrmion or for mumax3, over MCP pass `engine="mm2d"` or `engine="mumax3"`. **Run setup** (Deep mode) has an **Engine** menu; an engine that cannot run here (for example mumax3 without a GPU) is marked "not installed".

Engine	What it computes	Notes
macrospin (default)	Two coupled LLG equations for the sublattices m_1 and m_2: exchange A0, anisotropy Ku, Zeeman field, spin-orbit torque, thermal noise, time-dependent drives.	Built in; sweeps, continuation and combination of runs.

mm2d (built in)	SpinMate's own 2D micromagnetics of a ferromagnetic film one cell thick: exchange, perpendicular anisotropy, interfacial or bulk DMI, Zeeman and the local thin-film demagnetising field. Finds skyrmions, skyrmionium and domain walls, or runs LLG dynamics.	Part of SpinMate: CPU only, nothing to install; a 128 × 128 film relaxes in seconds.
mumax3 (optional plug-in)	Finite-difference micromagnetics of one ferromagnet on a grid, on the GPU. Input as SpinMate lines (turned into an `.mx3` script) or a complete `.mx3` script; the card shows ⟨m⟩(t).	Needs an NVIDIA GPU and the mumax3 binary in `MUMAX3_BIN`; without it validation warns. Single runs only.

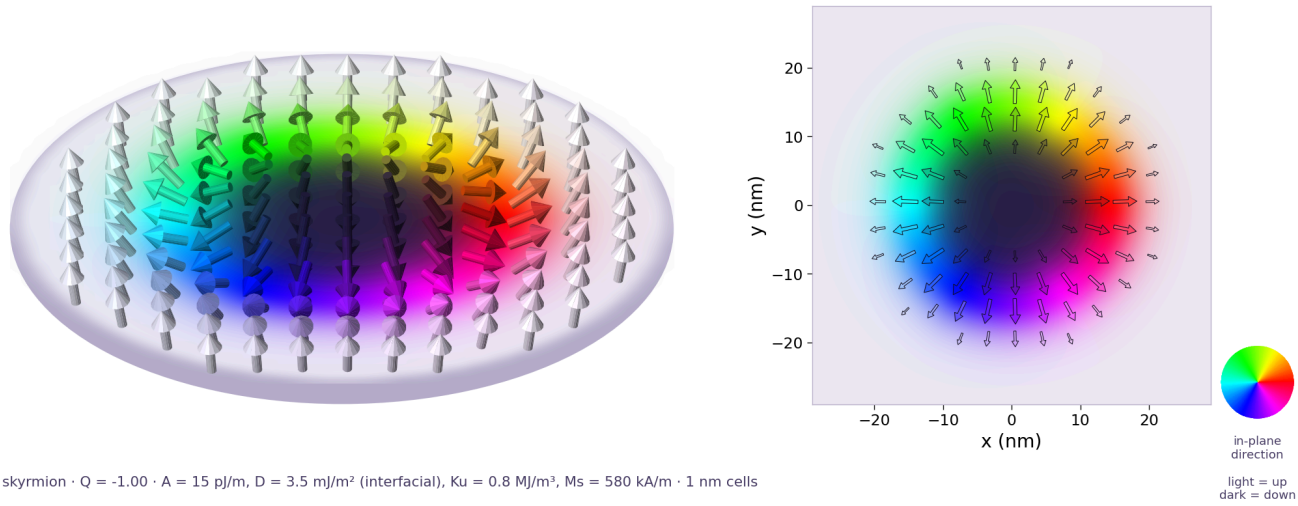
```
--Nx=128
--Ny=128
--Msat=800e3
--Aex=13e-12
--alpha=0.02
--B_ext=0,0,0.1
--t=2e-9
```

mumax3 is free software under the GNU GPL v3; SpinMate starts it as a separate program and stays MIT. If you publish results obtained with it, cite Vansteenkiste et al., *The design and verification of MuMax3*, AIP Advances 4, 107133 (2014). Outside Docker: `pip install -e plugins/spinmate-mumax3`.

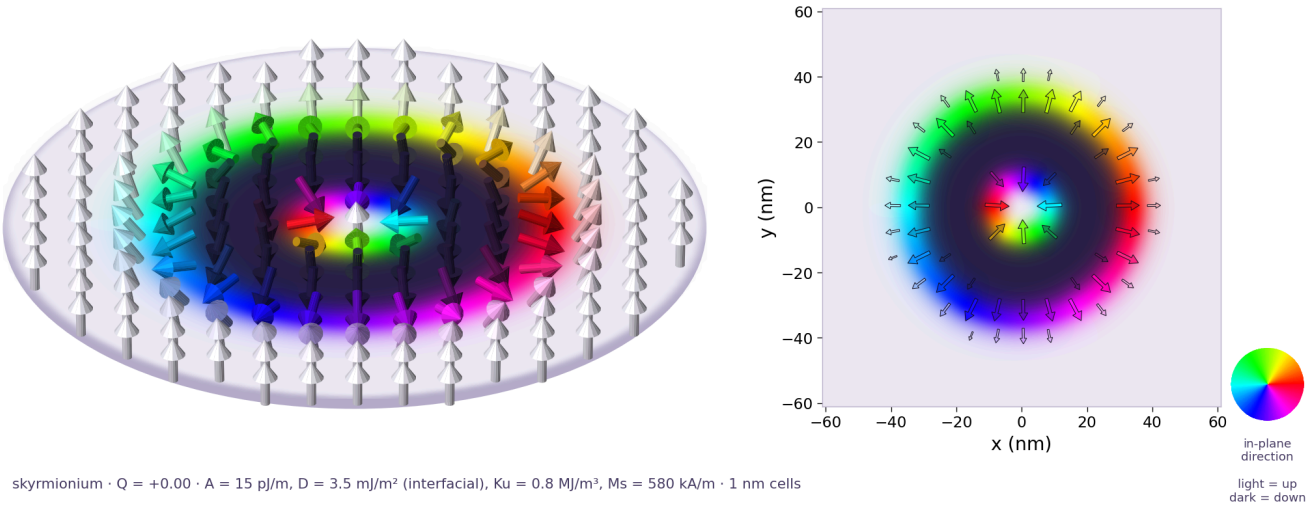
2D micromagnetics: skyrmions and spin waves with mm2d

Choose **mm2d** in the Engine menu of Run setup (Deep mode), or ask in the chat, for example “relax a Néel skyrmion with D = 3.5 mJ/m²” or “send 80 GHz spin waves from a stripe antenna”. `mode=relax` lowers the energy from a seed (`texture=skyrmion`, `skyrmionium`, `domain_wall`, `uniform`, `random`) to the nearest stable state; `mode=dynamics` relaxes first and then solves LLG in time, with an optional field pulse `B_pulse` for `t_pulse`, and reports the main frequency.

```
--mode=relax
--texture=skyrmion
--Aex=15e-12
--Ku=0.8e6
--Msat=580e3
--D=3.5e-3
--Nx=128
--Ny=128
```



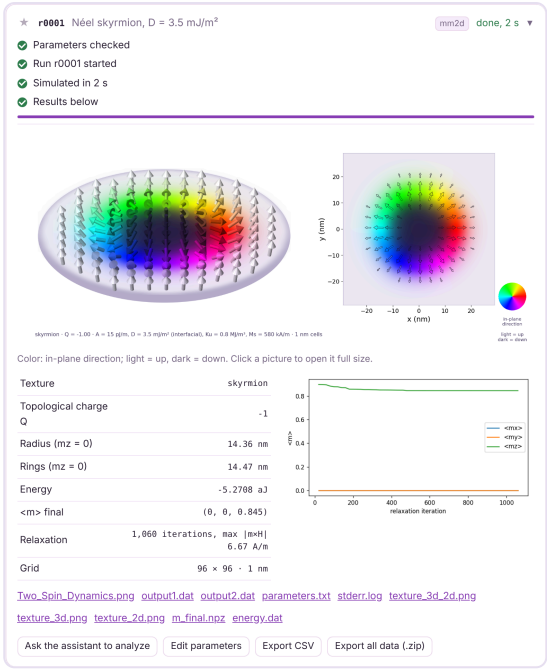
A relaxed Néel skyrmion: tilted view with 3D arrows (left) and top view with nm axes (right). Colour: in-plane direction; light = up, dark = down. Q = −1, radius 14.4 nm.



Skyrmionium, a skyrmion inside a skyrmion: Q = 0, rings at 8.5 nm and 34.4 nm.

The run card of an mm2d run shows the final texture in both views, then a table with the texture it found, the topological charge Q, the radius or the rings where $m_z = 0$, the energy and how far the relaxation went, and ⟨m⟩ during the relaxation or in time. `m_final.npz` keeps the full state; click a picture to open it full size.

mm2d check	Expected	Measured
Wall energy, D = 0 / 2 mJ/m²	$4\sqrt{(A \cdot Keff)} - \pi\lvert D\rvert$	within 0.3 %
Wall width	$\sqrt{(A/Keff)}$	within 1 %
Wall type	Néel (interfacial), Bloch (bulk)	as expected
Kittel frequency	$\gamma_o/2\pi \cdot (H + 2Keff/\mu_o Ms)$	within 0.2 %
Energy, α = 0	conserved	drift < 1e-4
Skyrmion / skyrmionium	Q = −1 / 0	−1.00 / 0.00
Spin wave, 80 GHz	λ = 49.6 nm	49.3 nm
Dispersion f(k)	analytic curve	within 1 %

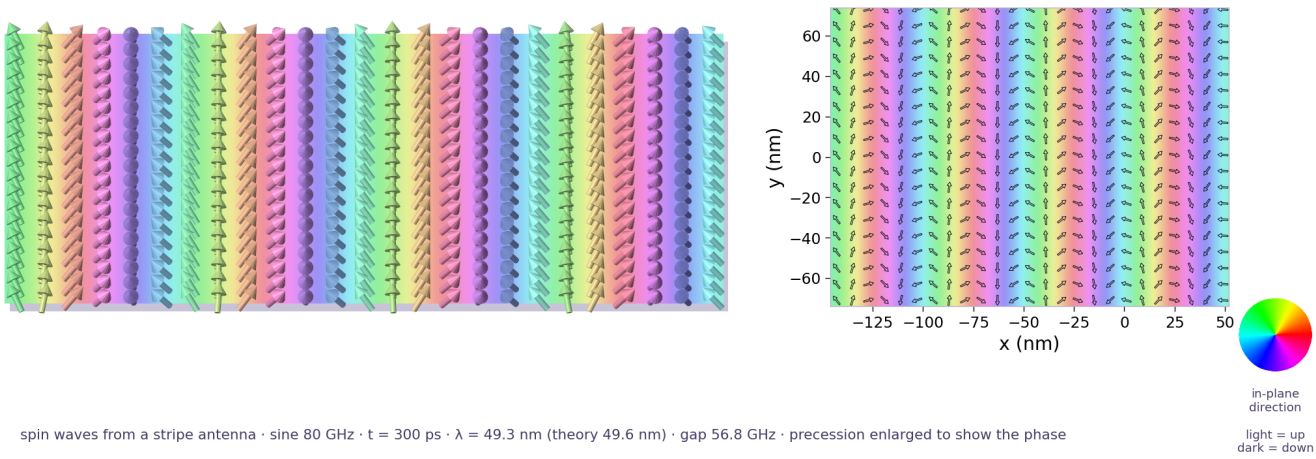


The run card of an mm2d skyrmion.

Spin waves

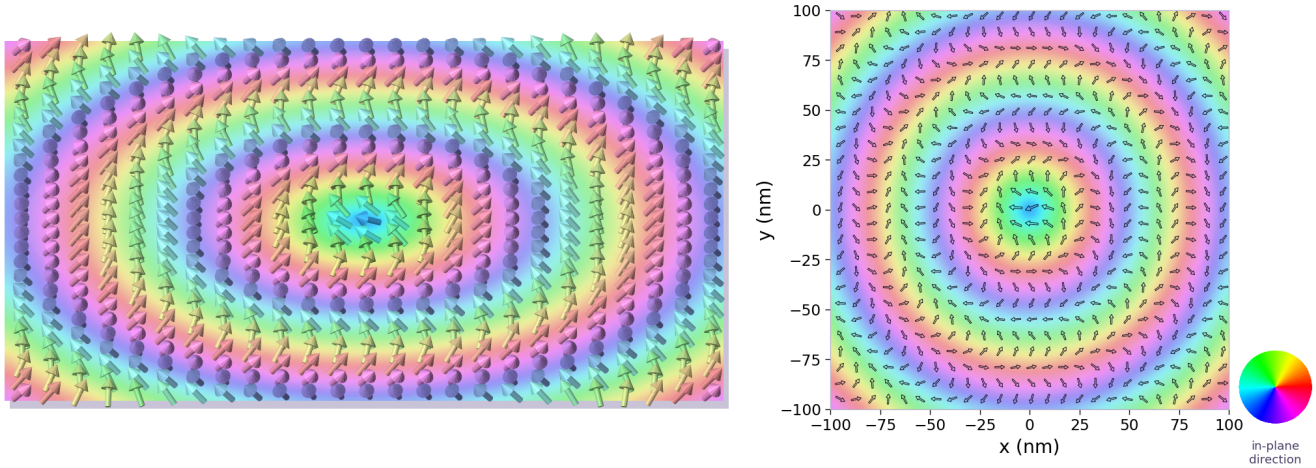
With `excite=stripe` or `excite=dot` the field pulse acts on an antenna instead of the whole film. `pulse=sine` at `f_rf` sends a monochromatic wave: the card gives its wavelength next to the theory. `pulse=sinc` up to `f_rf` excites every wave below that frequency at once and draws the dispersion f(k) with the analytic curve $f = \gamma_o/2\pi \cdot (H + 2K_{eff}/\mu_o M_s + 2A\,k^2/\mu_o M_s)$. Set `absorb` to about two wavelengths so the waves do not come back from the edges. Below the gap (here 56.8 GHz) nothing propagates.

```
--mode=dynamics
--texture=uniform
--Nx=300
--Ny=100
--dx=2e-9
--excite=stripe
--excite_x=-150e-9
--pulse=sine
--f_rf=80e9
--B_pulse=0.02,0,0
--t=300e-12
--t_pulse=300e-12
--table_dt=2e-12
--absorb=100e-9
```



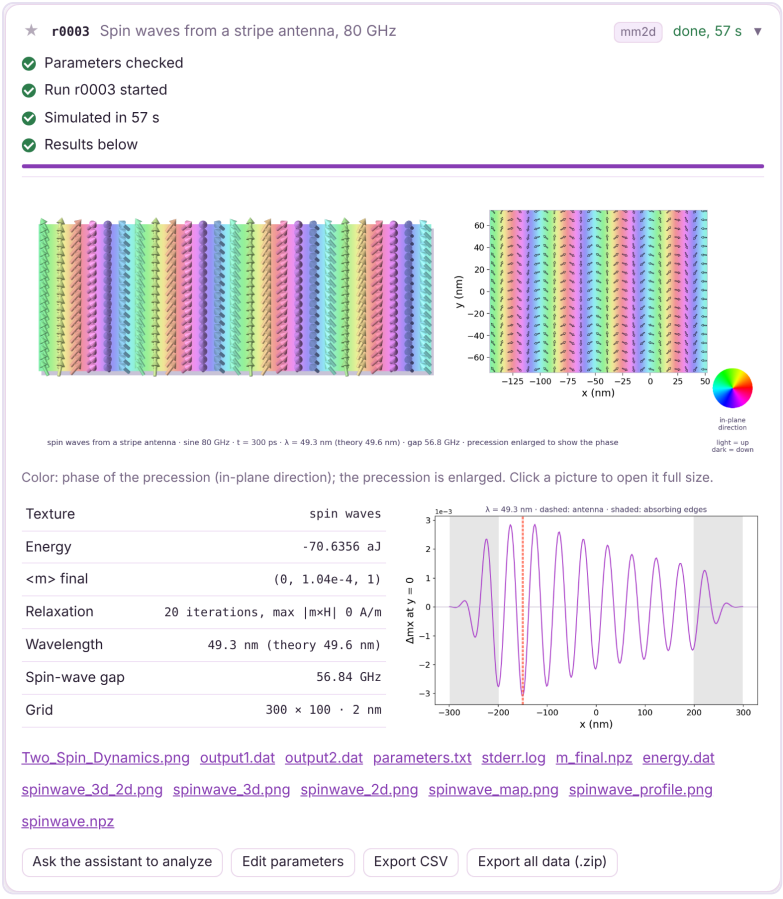
spin waves from a stripe antenna · sine 80 GHz · t = 300 ps · λ = 49.3 nm (theory 49.6 nm) · gap 56.8 GHz · precession enlarged to show the phase

Spin waves at 80 GHz from a stripe antenna, in the same views as a texture. Colour: phase of the precession; the precession is enlarged so it can be seen. λ = 49.3 nm (theory 49.6 nm).

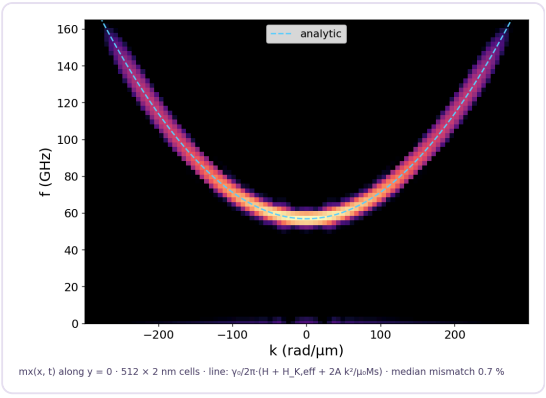


spin waves from a dot antenna · sine 80 GHz · t = 250 ps · λ = 48.4 nm (theory 49.6 nm) · gap 56.8 GHz · precession enlarged to show the phase

Circular spin waves from a dot antenna.



The run card of a spin-wave run: wavelength, gap and the profile along y = 0.



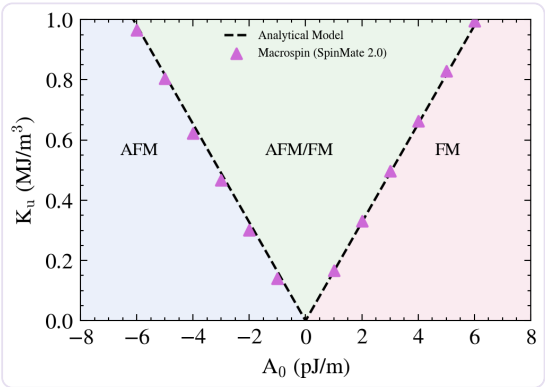
Dispersion from a sinc pulse up to 150 GHz; the dashed line is the analytic f(k).

Limits: the film is one cell thick and the demagnetising field is local ($-M_s \cdot m_z$), so there is no long-range stray field; no temperature yet. Two sublattices and a GPU version are planned.

10 Physics validation

`python -m spinmate_core.validation` runs the real solver on cases with an analytic answer and writes a report (agents read it as `spinmate://validation`). It runs on every pull request that changes the solver or the core. The shield in the left bar is green when every case passes; click it for the report and the phase diagram. Each sublattice carries Ms/2, so $\mu_0 H_E = 4|A0|/(Ms\ l^2)$ and $\mu_0 H_K = 4Ku/Ms$.

Case	Expected	Measured
$\lvert m_1\rvert = \lvert m_2\rvert = 1$	0	2e-16
Same input, same trajectory	0	0
Energy, α = 0, 0.5 ns	conserved	drift 1.6e-6
Kittel (FM mode)	5.62 GHz, 27.99 GHz/T	5.61 GHz, 28.03 GHz/T
AFM resonance	67.3 GHz	67.3 GHz
Time-step convergence	h = 1e-13 ≈ 2e-14 s	0.03 % apart



AFM/FM phase diagram: the boundary found by bisection with the solver lies on $|Ku| = 2|A0|/l^2$ within 2–6 %.

Gilbert damping, rate $\alpha\cdot\omega$	0.3525 /ns	0.3524 /ns
AFM/FM boundary	$\lvert Ku\rvert = 2\lvert A0\rvert/l^2$	as expected at ±10 %
Thermal field	isotropic, σ = Temp_Ampl	σ within 3 %

Fixed in 2.0. The first run found four issues in `macrospin/`: the static anisotropy was added twice (F1), the Ku(t) amplitude used g_o instead of μ_o (F4), the optional RK4 integrator had wrong weights (F2) and the thermal field was not Gaussian (F3). Runs at T = 0 without Ku(t) give bit-identical trajectories to version 1.0; runs with temperature change.

11 Sweeps, threshold searches and continuing runs

- **Sweep:** "Sweep A0 from −0.25e-12 to +0.25e-12 in 11 steps, 0.5 ns each." Runs go in parallel; the card shows a table and an overlay.
- **Form:** the chart icon next to the paperclip opens a sweep or threshold search form: parameter in convenient units, range, steps or what to look for, and duration. It writes the request with SI values and warns about unusual numbers.
- **Threshold search:** "Find the smallest SOT current between 50e6 and 150e6 A/cm² that makes the Néel vector rotate." A few rounds of bracketing sweeps narrow the interval.
- **Continue:** "Continue r0003 for another 0.5 ns." Time does not restart; *combine* joins the segments into one curve.

12 Comparing runs, figures and reports

- **Compare:** lists the parameters that differ and overlays the curves.
- **Figures:** "Make an IEEE figure of n_z for r0001 and r0003, first 60 ps." Styles science, IEEE, Nature; PDF, SVG and PNG downloads; *Change…* to adjust.
- **Reports:** SpinMate header, method, results, parameters, figures and appendix, with LaTeX math; print to PDF or download the Markdown version.

13 The Runs panel and questions across runs

The list icon opens every run: search by id, label, #tag or parameter; filter by status and end state; star and tag; select several to compare, plot, report or delete. With the catalog the assistant answers questions across runs in one read-only query:

> *Make me a table of all finished runs: id, A0, initial and final order, first n_z switch in ps.*

14 Settings and conversations

- **Language:** English (US or UK), Greek, Spanish, German or Italian; the buttons, Athena and the answers follow it. It starts from the browser language.
- **API keys:** under each key a check mark and the number of models it can use, or why it does not work. One working key is enough; a key that fails is not used. The × deletes a key from the tab (keys in `.env` are changed there).
- **Theme:** Spin (default), Light or Dark. **Default parameters:** saved per user and used as the base of every run.

- **Cost:** tokens and the estimated € next to the mode buttons; click it for **usage and cost** per day and per provider (Claude, OpenAI, Gemini).
- **Keyboard:** Ctrl+K new conversation, Ctrl+/ the Runs panel, Enter send, Shift+Enter new line, Esc closes a window (Cmd on a Mac).
- **Ask another god:** under an answer, sends the same question to another model level and shows both answers side by side.
- **Conversations:** the clock icon lists your chats; select one, several or all and delete them (runs are kept).
- **Voice:** the microphone records a request; the arrow next to it chooses the microphone and language.

15 Data, privacy and security

- The app listens on this computer only (`BIND_ADDRESS=127.0.0.1`). There is no password; expose it only behind a proxy with real authentication. The MCP endpoint `/mcp` uses the same port and the same rule.
- API keys stay in `.env` (never committed) or in the browser tab. Calls are billed to your own provider account; set `TOKEN_BUDGET` and a monthly limit in the provider console.
- Accounts (name and email) are stored only in `/data/accounts.json` on your machine; nothing is sent to the authors.
- The assistant reads the catalog with read-only SQL and a timeout, and starts runs only through the solver's parser. Over MCP, large requests run only after you agree (`confirmed=true`); in the chat only the **Run it** button approves them.

16 Configuration (`.env`)

Variable	Purpose
`ANTHROPIC_API_KEY` , `OPENAI_API_KEY` , `GEMINI_API_KEY`	Server-side keys (optional; can be pasted in Settings)
`HOST_PORT` , `BIND_ADDRESS`	Port (8421) and who may open the app (127.0.0.1)
`TOKEN_BUDGET`	Token limit per server session
`CONFIRM_MINUTES` , `CONFIRM_RUNS`	When a request waits for **Run it**
`MAX_PARALLEL` , `MAX_SWEEP_RUNS`	Parallel runs and the largest sweep
`DATABASE_URL` , `POSTGRES_PASSWORD`	Run catalog; empty URL switches it off
`EUR_PER_USD` , `PRICES_JSON`	Fixed exchange rate (default: daily ECB) and price overrides
`SMTP_HOST` , `SMTP_PORT` , `SMTP_USER` , `SMTP_PASSWORD` , `SMTP_FROM`	Mail server for the optional welcome email (Gmail: smtp.gmail.com, 587, App Password)
`ADMIN_TOKEN` , `ADMIN_SHEET_URL`	Turns on `/admin` (accounts, welcome emails, outbox, cost of this server) and links the SpinMate users Sheet
`APP_URL`	Base of the links in MCP results (default `http://localhost:8421`)
`MUMAX3_BIN`	Path of the mumax3 binary, for the mumax3 engine
`SSL_CERT_FILE`	Company CA bundle behind a corporate proxy

17 Parameters (most used)

Parameter	Meaning
`--t`	Duration (s)
`--A0`	Inter-sublattice exchange (J/m); negative couples antiparallel
`--m1`, `--m2`	Initial unit vectors
`--SOT_DC_Amp`, `--SOT_pol`	DC spin-orbit current (A/cm²) and its polarisation
`--flagTempVarying`, `--Temp`, `--Fr`	FeRh temperature cycle: on, amplitude (K), frequency (Hz)

A parameter file is plain text, one `--name=value` per line; attach it with the paper clip. Ask "Which parameters can I set?" for the full list with units. Examples are in `examples/`.

```
--t=5e-11
--A0=-0.248e-12
--m1=0.25,0.0,0.968
--m2=0.35,0.0,0.937
```

18 Troubleshooting

Symptom	What to do
"Add an API key"	Paste a key in Settings, or set it in `.env` and run `docker compose up -d --force-recreate`.
No answer, 401 / 429 errors	Check the key and the provider credit; on Auto the next model by score is tried.
New features do not appear	`git pull`, `docker compose up --build -d --force-recreate`, then Ctrl+Shift+R in the browser.
No welcome email	Check `SMTP_*` in `.env`; `/api/config` must show `"email":true`. The guide is always at `/api/guide.pdf`.
Page does not open from another computer	By design; see `BIND_ADDRESS` and chapter 15.
Claude Code does not see SpinMate	Start the Docker app, then `claude mcp list`; the server must show as connected. Restart Claude Code after `claude mcp add`.
"mumax3 not found"	Install mumax3 on a machine with an NVIDIA GPU and set `MUMAX3_BIN`; macrospin runs need nothing extra.
Microphone not listed	Allow the microphone once; open the app on localhost or https.

More in the README on GitHub. Issues and feedback are welcome.